

WASP-11b

R-1315

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_191218 · created 2026-07-31T18:59:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458835.8006 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.011
Transit depth (Rp/Rs)^2	1.23 +/- 0.24 [%]
Orbital Inclination (inc)	88.82 +/- 0.74
Airmass coefficient 1 (a1)	0.9952 +/- 0.0095
Airmass coefficient 2 (a2)	0.0027 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#3 - [378, 429]
Optimal Method	PSF photometry
Transit Duration (day)	0.1016 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.111 ± 0.011 vs 0.131 (deviation 1.82σ)
Duration fit vs published	0.1016 d vs 0.1075 d (deviation 1.44σ)
Mid-time O–C	-2.1 min
Depth SNR	10.1
Residual scatter	0.95 %

Light curve

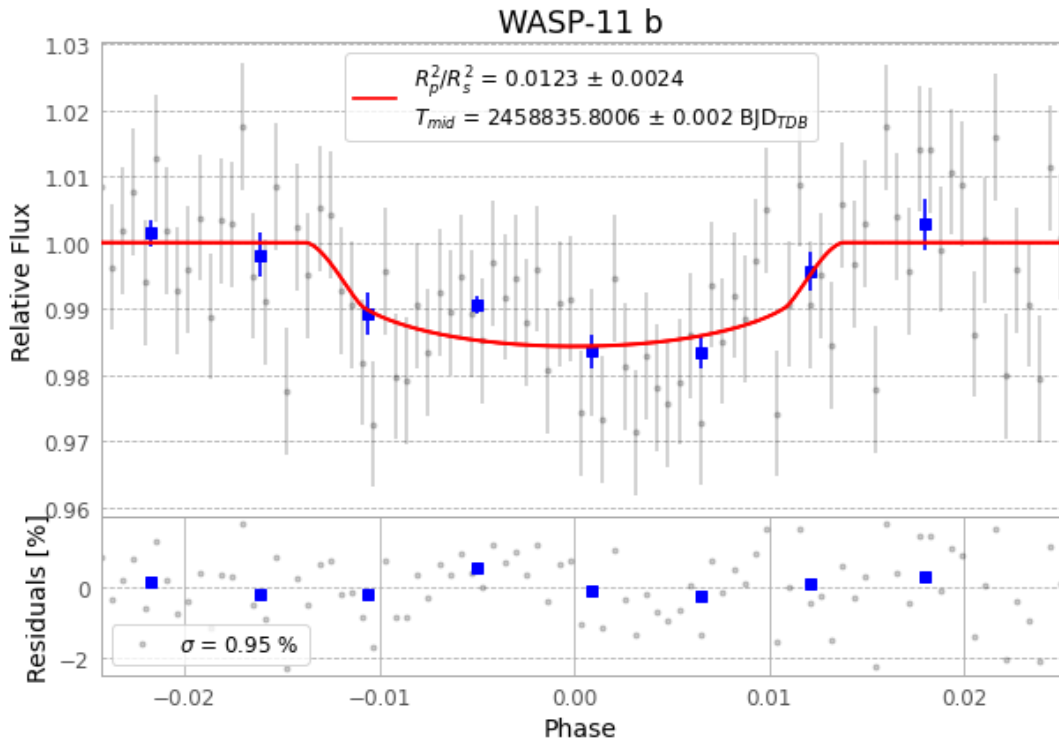


Figure 1 — FinalLightCurve_WASP-11 b_2019-12-17.png (SHA-256 3c10a5e1cf3350d9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 304], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b3a8af34757dd717...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), HATP-10191218045412.FITS → HATP-10191218091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-191218.log (bc44693747e1...), FinalLightCurve_WASP-11 b_2019-12-17.png (3c10a5e1cf33...), FinalLightCurve_WASP-11 b_2019-12-17.csv (1a46c0070a8e...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:59Z.